

HOMEWORK 5
ELEMENTARY LINEAR ALGEBRA
(MATH 2250-03), SPRING, 2018

CHECK: 03/15/2019

SECTION 2.8 - SUBSPACES OF $\mathbb{R}^n$

In problem 1 and problem 2, you are given a matrix A and an echelon form of A. Use this information to find a basis for $\text{Col}(A)$ and $\text{Nul}(A)$.

1. $A = \begin{bmatrix} 4 & 5 & 9 & -2 \\ 6 & 5 & 1 & 12 \\ 3 & 4 & 8 & -3 \end{bmatrix} \xrightarrow{\text{Row Reduce}} \begin{bmatrix} 1 & 2 & 6 & -5 \\ 0 & 1 & 5 & -6 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

Since the pivot columns of A form a basis for the column space of A (by Theorem 13 of section 2.8 in your textbook), the echelon form of A tells us that the first and second columns of A form a basis for $\text{Col}(A)$. That is, the set $\left\{ \begin{bmatrix} 4 \\ 6 \\ 3 \end{bmatrix}, \begin{bmatrix} 5 \\ 5 \\ 4 \end{bmatrix} \right\}$ is a basis for $\text{Col}(A)$.

To find a basis for $\text{Nul}(A)$, we begin by solving the system $Ax = 0$. Since the solution set to this system is preserved under elementary row operations, we use the echelon form of A to produce the system:

$$\begin{aligned} x_1 + 2x_2 + 6x_3 - 5x_4 &= 0 \\ x_2 + 5x_3 - 6x_4 &= 0 \\ 0 &= 0. \end{aligned}$$

By the usual method of back substitution, we solve this system and find the parametric vector form of the solution set:

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 4x_3 - 7x_4 \\ -5x_3 + 6x_4 \\ x_3 \\ x_4 \end{bmatrix} = x_3 \begin{bmatrix} 4 \\ -5 \\ 1 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} -7 \\ 6 \\ 0 \\ 1 \end{bmatrix}.$$

That is, the set $\left\{ \begin{bmatrix} 4 \\ -5 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -7 \\ 6 \\ 0 \\ 1 \end{bmatrix} \right\}$ is a basis for $\text{Nul}(A)$.

$$2. A = \begin{bmatrix} 1 & 4 & 8 & -3 & -7 \\ -1 & 2 & 7 & 3 & 4 \\ -2 & 2 & 9 & 5 & 5 \\ 3 & 6 & 9 & -5 & -2 \end{bmatrix} \xrightarrow{\text{Row Reduce}} \begin{bmatrix} 1 & 4 & 8 & 0 & 5 \\ 0 & 2 & 5 & 0 & -1 \\ 0 & 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Since the pivot columns of A form a basis for the column space of A (by Theorem 13 of section 2.8 in your textbook), the echelon form of A tells us that the first, second, and fourth columns of A

form a basis for $\text{Col}(A)$. That is, the set $\left\{ \begin{bmatrix} 1 \\ -1 \\ -2 \\ 3 \end{bmatrix}, \begin{bmatrix} 4 \\ 2 \\ 2 \\ 6 \end{bmatrix}, \begin{bmatrix} -3 \\ 3 \\ 5 \\ -5 \end{bmatrix} \right\}$ is a basis for $\text{Col}(A)$.

To find a basis for $\text{Nul}(A)$, we begin by solving the system $Ax = 0$. Since the solution set to this system is preserved under elementary row operations, we use the echelon form of A to produce the system:

$$\begin{aligned} x_1 + 4x_2 + 8x_3 + 5x_5 &= 0 \\ 2x_2 + 5x_3 - x_5 &= 0 \\ x_4 + 4x_5 &= 0 \\ 0 &= 0. \end{aligned}$$

By the usual method of back substitution, we solve this system and find the parametric vector form of the solution set:

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 2x_3 - 7x_5 \\ -(5/2)x_3 + (1/2)x_5 \\ x_3 \\ -4x_5 \\ x_5 \end{bmatrix} = x_3 \begin{bmatrix} 2 \\ -5/2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_5 \begin{bmatrix} -7 \\ 1/2 \\ 0 \\ -4 \\ 1 \end{bmatrix}.$$

That is, the set $\left\{ \begin{bmatrix} 2 \\ -5/2 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -7 \\ 1/2 \\ 0 \\ -4 \\ 1 \end{bmatrix} \right\}$ is a basis for $\text{Nul}(A)$.

SECTION 2.9 - DIMENSION AND RANK

3. Let $b_1 = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$ and $b_2 = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$. Then, $\mathcal{B} = \{b_1, b_2\}$ is a basis for the subspace of $\mathbb{R}^2$ spanned by $\mathcal{B}$. Find the vector x determined by the $\mathcal{B}$ -coordinate vector $[x]_{\mathcal{B}} = \begin{bmatrix} -1 \\ 3 \end{bmatrix}$.

Since the $\mathcal{B}$ -coordinate vector $[x]_{\mathcal{B}}$ tells us the coefficients in the linear combination of b_1 and b_2 which equals x , we have $x = (-1)b_1 + (3)b_2$. So, expanding, we reach our final answer:

$$x = (-1) \begin{bmatrix} -2 \\ 1 \end{bmatrix} + (3) \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 2+9 \\ -1+3 \end{bmatrix} = \begin{bmatrix} 11 \\ 2 \end{bmatrix}.$$

4. Let $\mathcal{B} = \{b_1, b_2\}$ with $b_1 = \begin{bmatrix} -3 \\ 1 \\ -4 \end{bmatrix}$ and $b_2 = \begin{bmatrix} 7 \\ 5 \\ -6 \end{bmatrix}$. You may assume that $x = \begin{bmatrix} 11 \\ 0 \\ 7 \end{bmatrix}$ is in the subspace of $\mathbb{R}^3$ spanned by $\mathcal{B}$, and you may also assume that $\mathcal{B}$ is a basis for this subspace. Find $[x]_{\mathcal{B}}$, the $\mathcal{B}$ -coordinate vector of x .

To find the $\mathcal{B}$ -coordinate vector of x , we must find the coefficients c_1 and c_2 such that $x = c_1 b_1 + c_2 b_2$, since $[x]_{\mathcal{B}} = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}$. To do so, observe that this simply reduces to solving the inhomogeneous linear system $\begin{bmatrix} b_1 & b_2 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = x$. That is, we need only solve

$$\begin{bmatrix} -3 & 7 \\ 1 & 5 \\ -4 & -6 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 11 \\ 0 \\ 7 \end{bmatrix}.$$

Reducing the augmented matrix $\begin{bmatrix} b_1 & b_2 & x \end{bmatrix}$ associated to this system, we produce

$$\begin{bmatrix} b_1 & b_2 & x \end{bmatrix} = \begin{bmatrix} -3 & 7 & 11 \\ 1 & 5 & 0 \\ -4 & -6 & 7 \end{bmatrix} \xrightarrow{\text{Row Reduce}} \begin{bmatrix} 1 & 0 & -5/2 \\ 0 & 1 & 1/2 \\ 0 & 0 & 0 \end{bmatrix}.$$

So, $c_1 = -5/2$ and $c_2 = 1/2$. Therefore, we reach our final solution: $[x]_{\mathcal{B}} = \begin{bmatrix} -5/2 \\ 1/2 \end{bmatrix}$.

SECTION 3.1 - INTRODUCTION TO DETERMINANTS

5. Let $A = \begin{bmatrix} 1 & -2 & 5 & 2 \\ 0 & 0 & 3 & 0 \\ 2 & -4 & -3 & 5 \\ 2 & 0 & 3 & 5 \end{bmatrix}$. Compute $\det(A)$.

Expanding across the second row for convenience (since it has only one entry not equal to zero), we produce

$$\begin{aligned} \det(A) &= -(0)\det\left(\begin{bmatrix} -2 & 5 & 2 \\ -4 & -3 & 5 \\ 0 & 3 & 5 \end{bmatrix}\right) + (0)\det\left(\begin{bmatrix} 1 & 5 & 2 \\ 2 & -3 & 5 \\ 2 & 3 & 5 \end{bmatrix}\right) \\ &\quad - (3)\det\left(\begin{bmatrix} 1 & -2 & 2 \\ 2 & -4 & 5 \\ 2 & 0 & 5 \end{bmatrix}\right) + (0)\det\left(\begin{bmatrix} 1 & -2 & 5 \\ 2 & -4 & -3 \\ 2 & 0 & 3 \end{bmatrix}\right) \\ &= -3\left((1)\det\left(\begin{bmatrix} -4 & 5 \\ 0 & 5 \end{bmatrix}\right) - (-2)\det\left(\begin{bmatrix} 2 & 5 \\ 2 & 5 \end{bmatrix}\right) + (2)\det\left(\begin{bmatrix} 2 & -4 \\ 2 & 0 \end{bmatrix}\right)\right) \\ &= -3\left(1((-4)(5) - (5)(0)) + 2((2)(5) - (5)(2)) + 2((2)(0) - (-4)(2))\right) \\ &= -3(-20 + 0 + 16) \\ &= -3(-4) \\ &= 12. \end{aligned}$$

6. Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and let $k \in \mathbb{R}$ be a scalar. Find a formula which relates $\det(kA)$ to $\det(A)$.

By direct computation, since

$$\det(A) = \det \left(\begin{bmatrix} a & b \\ c & d \end{bmatrix} \right) = ad - cb$$

and since

$$\det(kA) = \det \left(\begin{bmatrix} ka & kb \\ kc & kd \end{bmatrix} \right) = (ka)(kd) - (kb)(kc) = k^2(ad - bc)$$

we may conclude that $\det(kA) = k^2 \det(A)$.

SECTION 3.2 - PROPERTIES OF DETERMINANTS

7. Use determinants to decide if the set of vectors $\left\{ \begin{bmatrix} 4 \\ 6 \\ 2 \end{bmatrix}, \begin{bmatrix} -7 \\ 0 \\ 7 \end{bmatrix}, \begin{bmatrix} -3 \\ -5 \\ -2 \end{bmatrix} \right\}$ is linearly independent.

Recall that the Invertible Matrix Theorem tells us that a set of n vectors $\{v_1, \dots, v_n\} \subseteq \mathbb{R}^n$ is linearly independent if and only if the matrix $A = \begin{bmatrix} v_1 & \dots & v_n \end{bmatrix}$ (that is, the matrix with $\{v_1, \dots, v_n\}$ as its columns) is invertible. So, we need only determine if the matrix $\begin{bmatrix} 4 & -7 & -3 \\ 6 & 0 & -5 \\ 2 & 7 & -2 \end{bmatrix}$ is invertible

to determine if $\left\{ \begin{bmatrix} 4 \\ 6 \\ 2 \end{bmatrix}, \begin{bmatrix} -7 \\ 0 \\ 7 \end{bmatrix}, \begin{bmatrix} -3 \\ -5 \\ -2 \end{bmatrix} \right\}$ is linearly independent. To that end, recall that a square matrix A is invertible if and only if $\det(A) \neq 0$ (this is Theorem 4 in Section 3.2 of your textbook). Hence, we carry out the following computation:

$$\begin{aligned} \det \left(\begin{bmatrix} 4 & -7 & -3 \\ 6 & 0 & -5 \\ 2 & 7 & -2 \end{bmatrix} \right) &= (4)\det \left(\begin{bmatrix} 0 & -5 \\ 7 & -2 \end{bmatrix} \right) - (-7)\det \left(\begin{bmatrix} 6 & -5 \\ 2 & -2 \end{bmatrix} \right) + (-3)\det \left(\begin{bmatrix} 6 & 0 \\ 2 & 7 \end{bmatrix} \right) \\ &= 4(35) + 7(-2) - 3(42) \\ &= 0. \end{aligned}$$

Therefore, we may conclude $\left\{ \begin{bmatrix} 4 \\ 6 \\ 2 \end{bmatrix}, \begin{bmatrix} -7 \\ 0 \\ 7 \end{bmatrix}, \begin{bmatrix} -3 \\ -5 \\ -2 \end{bmatrix} \right\}$ is not linearly independent.

8. Let A be an invertible $n \times n$ matrix. Prove that $\det(A^{-1}) = \frac{1}{\det(A)}$.

Recall that if A and B are two $n \times n$ matrices, then $\det(AB) = (\det(A))(\det(B))$ (this is Theorem 6 of section 3.2 in your textbook). So, if A is an invertible matrix, then we have

$$\det(I) = \det(AA^{-1}) = (\det(A))(\det(A^{-1})).$$

Recall now that the determinant of a triangular matrix is the product of the entries on the main diagonal of the matrix (by Theorem 2 of Section 3.1 in your textbook). So, since I is diagonal, it is certainly triangular, and we have $\det(I) = 1$. But notice, this then says that

$$(\det(A))(\det(A^{-1})) = 1.$$

Therefore, we reach our final answer and conclude

$$\det(A) = \frac{1}{\det(A^{-1})}, \quad \text{or, equivalently,} \quad \det(A^{-1}) = \frac{1}{\det(A)}.$$

As a side note, observe that if we write $\frac{1}{a}$ as a^{-1} for some real number $a \in \mathbb{R}$ in the usual fashion, the above formula then says

$$\det(A^{-1}) = (\det(A))^{-1}.$$

That is, inverses *commute with the determinant* (meaning that we can always pull the inverse outside the determinant).

SECTION 3.3 - CRAMER'S RULE, VOLUME, AND LINEAR TRANSFORMATIONS

9. Compute the adjugate of $A = \begin{bmatrix} 1 & 1 & 3 \\ -2 & 2 & 1 \\ 0 & 1 & 1 \end{bmatrix}$. Then, using the adjugate, compute A^{-1} .

The nine cofactors of A are:

$$\begin{aligned} C_{11} &= (1)\det\left(\begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}\right) = 1, & C_{12} &= (-1)\det\left(\begin{bmatrix} -2 & 1 \\ 0 & 1 \end{bmatrix}\right) = 2, & C_{13} &= (1)\det\left(\begin{bmatrix} -2 & 2 \\ 0 & 1 \end{bmatrix}\right) = -2 \\ C_{21} &= (-1)\det\left(\begin{bmatrix} 1 & 3 \\ 1 & 1 \end{bmatrix}\right) = 2, & C_{22} &= (1)\det\left(\begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}\right) = 1, & C_{23} &= (-1)\det\left(\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}\right) = -1 \\ C_{31} &= (1)\det\left(\begin{bmatrix} 1 & 3 \\ 2 & 1 \end{bmatrix}\right) = -5, & C_{32} &= (-1)\det\left(\begin{bmatrix} 1 & 3 \\ -2 & 1 \end{bmatrix}\right) = -7, & C_{33} &= (1)\det\left(\begin{bmatrix} 1 & 1 \\ -2 & 2 \end{bmatrix}\right) = 4 \end{aligned}$$

So, by definition, we have

$$\text{adj}(A) = \begin{bmatrix} C_{11} & C_{12} & C_{13} \\ C_{21} & C_{22} & C_{23} \\ C_{31} & C_{32} & C_{33} \end{bmatrix}^T = \begin{bmatrix} 1 & -2 & -2 \\ -2 & 1 & 1 \\ -5 & 7 & 4 \end{bmatrix}^T.$$

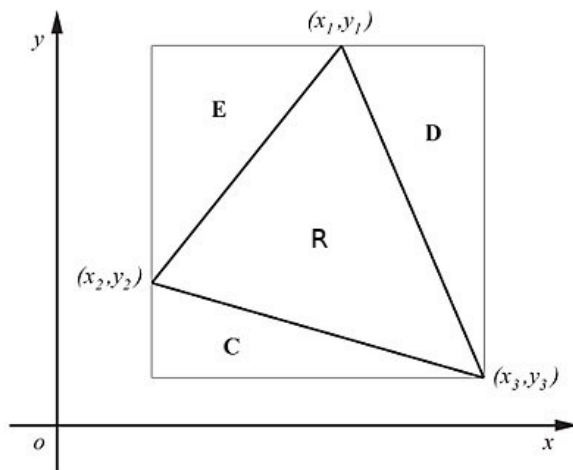
Since $A^{-1} = \frac{1}{\det(A)}\text{adj}(A)$ (this is Theorem 8 in Section 3.3 of your textbook), we then have that

$$A^{-1} = \frac{1}{\det(A)}\text{adj}(A) = \frac{1}{-3} \begin{bmatrix} 1 & 2 & -2 \\ 2 & 1 & -1 \\ -5 & -7 & 4 \end{bmatrix}^T = \begin{bmatrix} -1/3 & -2/3 & 5/3 \\ -2/3 & -1/3 & 7/3 \\ 2/3 & 1/3 & -4/3 \end{bmatrix}.$$

10. Let (x_1, y_1) , (x_2, y_2) and (x_3, y_3) be points in $\mathbb{R}^2$, and let R be the triangular region in $\mathbb{R}^2$ whose vertices correspond to these points. Write $A(R)$ to denote the area of R . Show that

$$A(R) = \frac{1}{2} \det \begin{pmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{pmatrix}.$$

By Theorem 9 of Section 3.3 in your textbook, we know that the volume of the parallelogram determined by the columns of a 2×2 matrix A is $|\det(A)|$. Drawing a box around our triangle and labeling the areas of the box which are not part of the triangle, we have something like this:



Clearly, the area of R is just the area of the rectangle (let's call it Q) minus the areas of the triangles E , D , and C . That is,

$$A(R) = A(Q) - A(E) - A(D) - A(C).$$

By inspection, we can see that

$$A(Q) = (x_3 - x_2)(y_1 - y_3)$$

$$A(C) = 1/2(x_3 - x_2)(y_2 - y_3)$$

$$A(E) = 1/2(x_1 - x_2)(y_1 - y_2)$$

$$A(D) = 1/2(x_3 - x_1)(y_1 - y_3)$$

So, we have

$$A(R) = (x_3 - x_2)(y_1 - y_3) - \frac{1}{2}((x_3 - x_2)(y_2 - y_3) + (x_1 - x_2)(y_1 - y_2) + (x_3 - x_1)(y_1 - y_3)).$$

Collecting terms and rearranging, this yields

$$A(R) = \frac{1}{2}((x_2y_3 - x_3y_2) - (x_1y_3 - x_3y_1) + (x_1y_2 - x_2y_1))$$

. Computing the determinant in the statement of the problem, we have

$$\begin{aligned} \frac{1}{2} \det \begin{pmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{pmatrix} &= \frac{1}{2} (x_1(y_2 - y_3) - y_1(x_2 - x_3) + 1(x_2y_3 - y_2x_3)) \\ &= \frac{1}{2} (x_1y_2 - x_1y_3 - y_1x_2 + y_1x_3 + x_2y_3 - y_2x_3) \\ &= \frac{1}{2} ((x_2y_3 - x_3y_2) - (x_1y_3 - x_3y_1) + (x_1y_2 - x_2y_1)) = A(R). \end{aligned}$$